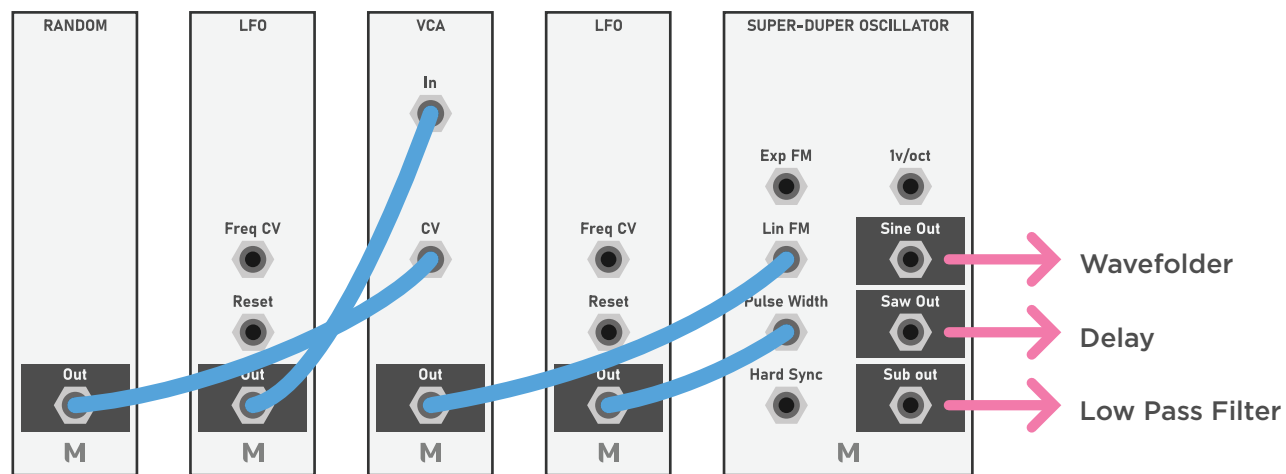


# Drones

## Oscillators

- Audio
- Control Voltage
- Trigger / Gate

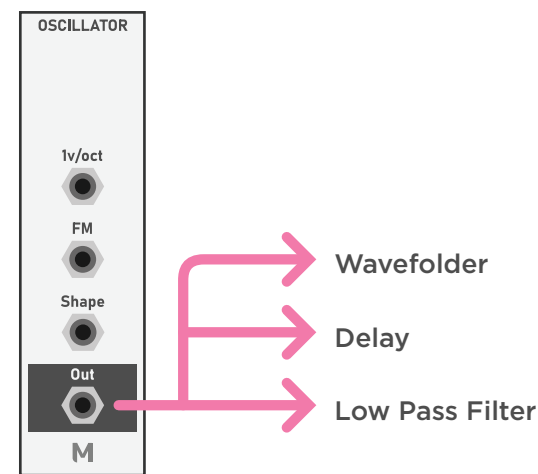
# MONOTRAIL TECH TALK



(a)

In general, the more in- and outputs your oscillator has, the more options you have for patching and sound modulation. On the input side, things like exponential and linear frequency modulation, pulse width control, and a sync input, will give you ways to change the character of a sound over time.

a - For example, take a slow LFO to modulate the pulse-width input. And feed an audio rate LFO through a VCA, opened by a slow random voltage, to modulate the linear FM input. Multiple un-synced modulation sources to different parameters will keep things dynamic.



(b)

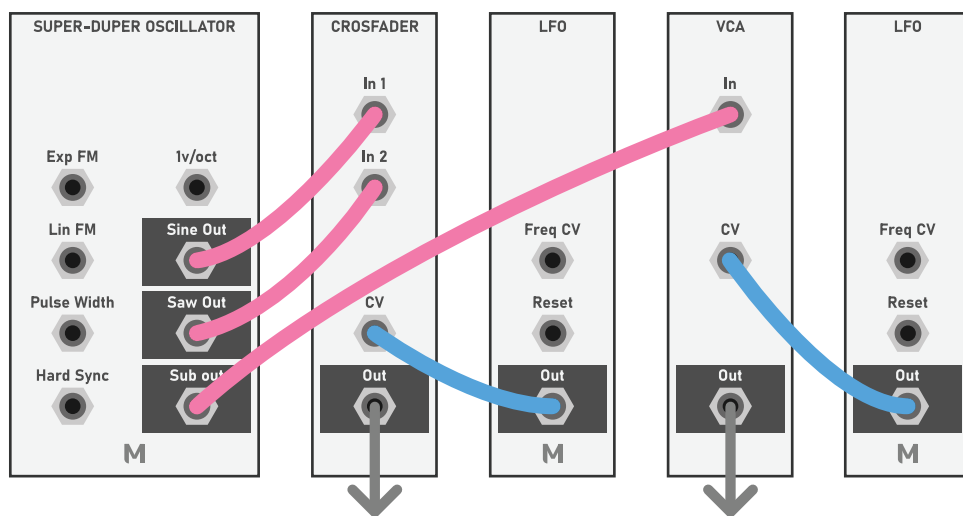
If you have multiple outputs on your oscillator, you can use those to create different audio paths.

b - Even if you only have one output, it's still worth it to multiply that output into paths with different treatments.

# Drones

## Oscillators

- Audio
- Control Voltage
- Trigger / Gate



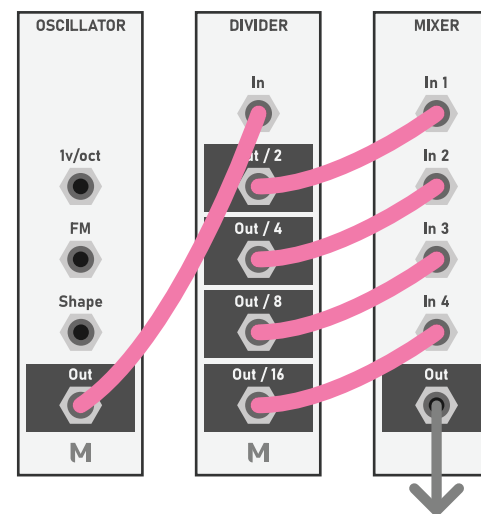
(a)

On the output side, simultaneous different waveform outputs and/or a sub output, will give you more signal paths to mess around with. It works great to send the different outputs through different effects and patches, and then bring them back together in a later stage.

a - A simple trick is to send two different waveforms to a crossfader, to give you some basic wave shape morphing. It can also be worth it to try and send audio through a VCA, and then modulate that VCA with an LFO at audio rates.

# MONOTRAIL

## TECH TALK



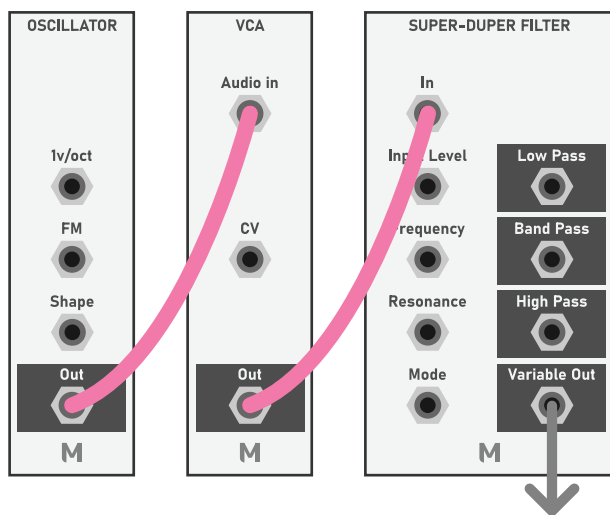
(b)

b - If your oscillator doesn't have a sub output, you can make one with an analog clock divider. Depending on the divisions it offers, you can even create chords this way.

# Drones

## Filters

— Audio  
— Control Voltage  
— Trigger / Gate

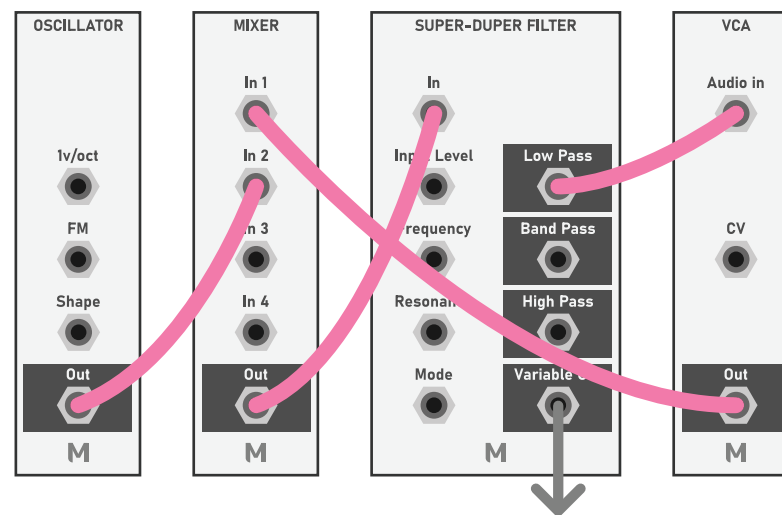


(a)

a - Multiple CV inputs and audio outputs are useful on filters as well. A simple trick with a filter that nicely overdrives, but does not have a CV input, is to patch a VCA between the Oscillator and the filter. This way, if you send something like an LFO to that VCA, you can create subtle movement.

# MONOTRAIL

## TECH TALK

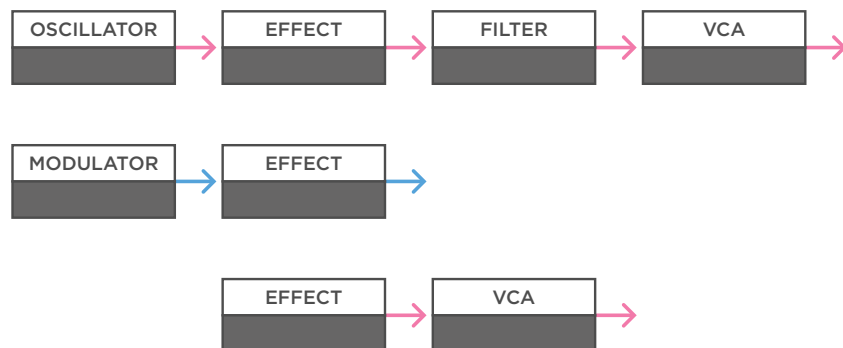
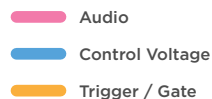


(b)

b - Similarly, if there's no CV input for resonance on your filter, you can try and see what happens if you patch up a feedback circuit yourself. For that you need to multiply the output of your filter, or use two different audio outputs of the same filter. One signal is to listen to. The other has to be passed through a VCA and then mixed with the signal you are originally feeding the filter. Now you can send something like a smooth random voltage to the VCA to slowly change the amount of feedback over time.

# Drones

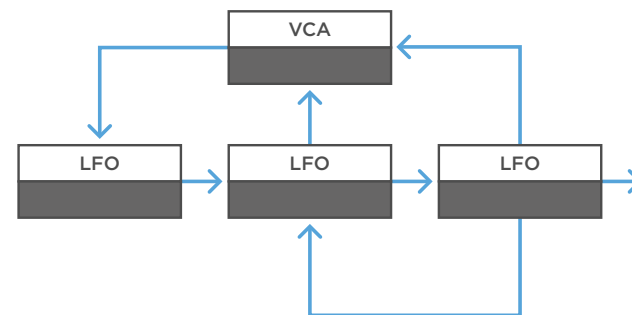
## Effects / Cross-modulation



(a)

a - It's a good idea to experiment with sound shaping and effect modules like wave-folders, delays, and reverbs in different places of your signal path. You don't have to use these at the end of your chain, you can experiment with them. For example, try running an oscillator through a delay or reverb before the filter and VCA. Or run modulations sources like envelopes and LFOs through distortions, wave-folders, or delays. You can also try to run outputs of effects modules through other effects, or through a VCA to chop them up. Play around with it!

# MONOTRAIL TECH TALK



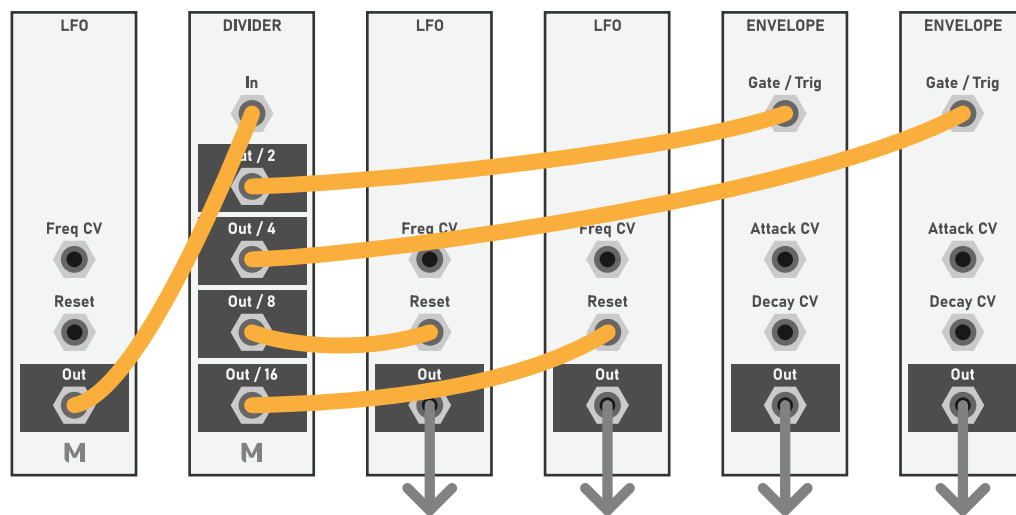
(b)

b - You can create nice fluctuating voltages by cross-modulating triangle or sinewave LFOs. I find that two LFOs quickly end up generating a loop. But if you add another LFO for triple cross-modulation, or have the amount of modulation between two LFOs changed by passing it through a VCA and control that VCA with a third LFO, you can add variation and make the signal as complex as you like.

# Drones

## Modulation

- Audio
- Control Voltage
- Trigger / Gate

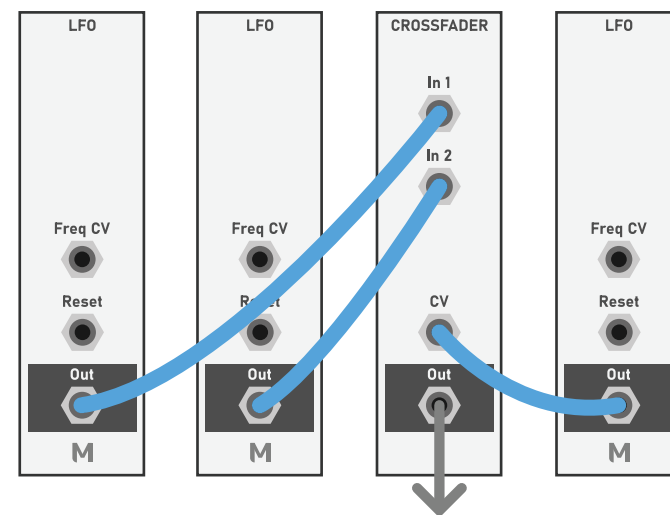


(a)

a - Here's an interesting patch for slow rhythmic drones. Use a clock source, like an LFO, and send it to a clock divider. Then use the different clock divisions to retrigger LFOs and trigger envelopes. Send all of them to different destinations to create slow related rhythmic modulation.

# MONOTRAIL

## TECH TALK



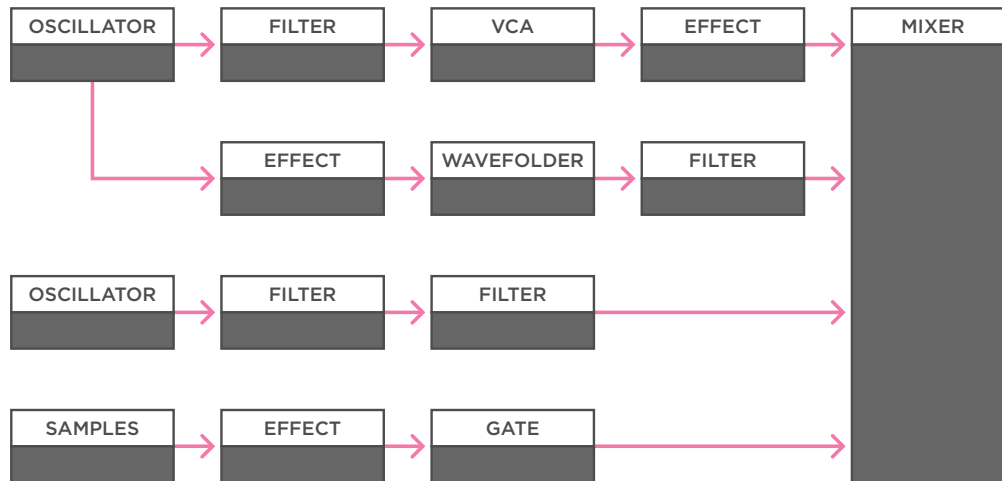
(b)

b - Using a cross-fader to fade between two modulation sources can give dynamic results. And like we covered in my previous video "Modular Utilities", if you don't have one, you can patch it up yourself with some basic utility, or make a patch where you fade one source, to two different destinations.

# Drones

## Mixing

— Audio  
— Control Voltage  
— Trigger / Gate

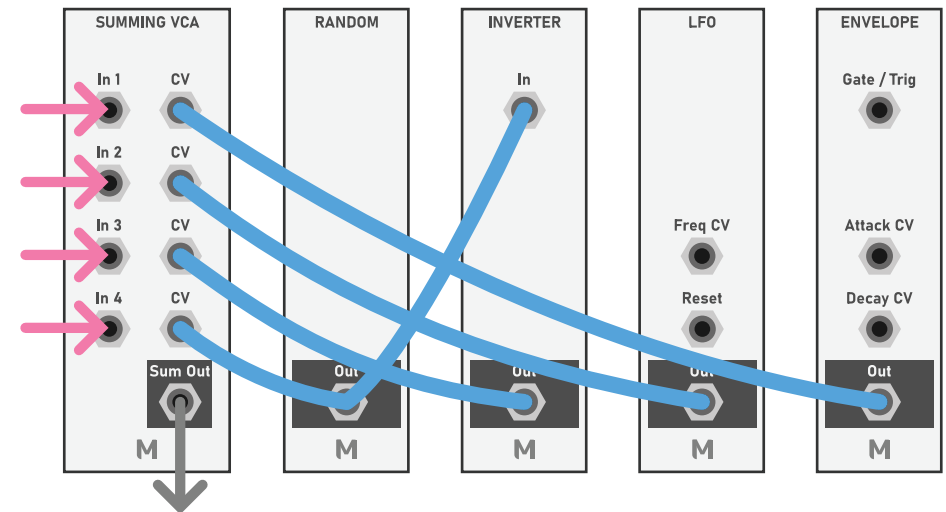


(a)

a - It works great to create different audio-paths from a single or multiple oscillators, and then mix them back together. Here you see an example with two oscillators and a sample player, resulting into four audio signals. Using a simple mixer to manually mix the different sounds you've created in and out can be enough to create a long drone piece that slowly changes over time.

# MONOTRAIL

## TECH TALK



(b)

b - You can also use summing VCA's to automate the mixing process as part of your drone. You can feed it different voices, or variations, and then use random voltages or complex LFO's to change their volume over time. It works great to multiply a modulation source (in the illustration the random voltage) and invert one of them, to have one sound oppressed, while another rises. Using different un-synced modulation will give more organic shifting changes.